

4.4 Constrained Optimization

Sometimes we wish not only to maximize or minimize a function $f(\mathbf{x})$ over all possible values of $\mathbf{x}$. Instead we may wish to find the maximal or minimal value of $f(\mathbf{x})$ for values of $\mathbf{x}$ in some set $\mathbb{S}$. This is known as **constrained optimization**. Points $\mathbf{x}$ that lie within the set $\mathbb{S}$ are called **feasible** points in constrained optimization terminology.

We often wish to find a solution that is small in some sense. A common approach in such situations is to impose a norm constraint, such as $\|\mathbf{x}\| \leq 1$.

One simple approach to constrained optimization is simply to modify gradient descent taking the constraint into account. If we use a small constant step size ϵ , we can make gradient descent steps, then project the result back into $\mathbb{S}$. If we use a line search, we can search only over step sizes ϵ that yield new $\mathbf{x}$ points that are feasible, or we can project each point on the line back into the constraint region. When possible, this method can be made more efficient by projecting the gradient into the tangent space of the feasible region before taking the step or beginning the line search ([Rosen, 1960](#)).

A more sophisticated approach is to design a different, unconstrained optimization problem whose solution can be converted into a solution to the original, constrained optimization problem. For example, if we want to minimize $f(\mathbf{x})$ for

$\mathbf{x} \in \mathbb{R}^2$ with $\mathbf{x}$ constrained to have exactly unit L^2 norm, we can instead minimize $g(\theta) = f([\cos \theta, \sin \theta]^\top)$ with respect to θ , then return $[\cos \theta, \sin \theta]$ as the solution to the original problem. This approach requires creativity; the transformation between optimization problems must be designed specifically for each case we encounter.

The **Karush–Kuhn–Tucker** (KKT) approach¹ provides a very general solution to constrained optimization. With the KKT approach, we introduce a new function called the **generalized Lagrangian** or **generalized Lagrange function**.

To define the Lagrangian, we first need to describe $\mathbb{S}$ in terms of equations and inequalities. We want a description of $\mathbb{S}$ in terms of m functions $g^{(i)}$ and n functions $h^{(j)}$ so that $\mathbb{S} = \{\mathbf{x} \mid \forall i, g^{(i)}(\mathbf{x}) = 0 \text{ and } \forall j, h^{(j)}(\mathbf{x}) \leq 0\}$. The equations involving $g^{(i)}$ are called the **equality constraints**, and the inequalities involving $h^{(j)}$ are called **inequality constraints**.

We introduce new variables λ_i and α_j for each constraint, these are called the KKT multipliers. The generalized Lagrangian is then defined as

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\alpha}) = f(\mathbf{x}) + \sum_i \lambda_i g^{(i)}(\mathbf{x}) + \sum_j \alpha_j h^{(j)}(\mathbf{x}). \quad (4.14)$$

We can now solve a constrained minimization problem using unconstrained optimization of the generalized Lagrangian. As long as at least one feasible point exists and $f(\mathbf{x})$ is not permitted to have value ∞ , then

$$\min_{\mathbf{x}} \max_{\boldsymbol{\lambda}} \max_{\boldsymbol{\alpha}, \boldsymbol{\alpha} \geq 0} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\alpha}) \quad (4.15)$$

has the same optimal objective function value and set of optimal points $\mathbf{x}$ as

$$\min_{\mathbf{x} \in \mathbb{S}} f(\mathbf{x}). \quad (4.16)$$

This follows because any time the constraints are satisfied,

$$\max_{\boldsymbol{\lambda}} \max_{\boldsymbol{\alpha}, \boldsymbol{\alpha} \geq 0} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\alpha}) = f(\mathbf{x}), \quad (4.17)$$

while any time a constraint is violated,

$$\max_{\boldsymbol{\lambda}} \max_{\boldsymbol{\alpha}, \boldsymbol{\alpha} \geq 0} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\alpha}) = \infty. \quad (4.18)$$

These properties guarantee that no infeasible point can be optimal, and that the optimum within the feasible points is unchanged.

¹The KKT approach generalizes the method of **Lagrange multipliers**, which allows equality constraints but not inequality constraints.

To perform constrained maximization, we can construct the generalized Lagrange function of $-f(\mathbf{x})$, which leads to this optimization problem:

$$\min_{\mathbf{x}} \max_{\boldsymbol{\lambda}} \max_{\boldsymbol{\alpha}, \boldsymbol{\alpha} \geq 0} -f(\mathbf{x}) + \sum_i \lambda_i g^{(i)}(\mathbf{x}) + \sum_j \alpha_j h^{(j)}(\mathbf{x}). \quad (4.19)$$

We may also convert this to a problem with maximization in the outer loop:

$$\max_{\mathbf{x}} \min_{\boldsymbol{\lambda}} \min_{\boldsymbol{\alpha}, \boldsymbol{\alpha} \geq 0} f(\mathbf{x}) + \sum_i \lambda_i g^{(i)}(\mathbf{x}) - \sum_j \alpha_j h^{(j)}(\mathbf{x}). \quad (4.20)$$

The sign of the term for the equality constraints does not matter; we may define it with addition or subtraction as we wish, because the optimization is free to choose any sign for each λ_i .

The inequality constraints are particularly interesting. We say that a constraint $h^{(i)}(\mathbf{x})$ is **active** if $h^{(i)}(\mathbf{x}^*) = 0$. If a constraint is not active, then the solution to the problem found using that constraint would remain at least a local solution if that constraint were removed. It is possible that an inactive constraint excludes other solutions. For example, a convex problem with an entire region of globally optimal points (a wide, flat region of equal cost) could have a subset of this region eliminated by constraints, or a nonconvex problem could have better local stationary points excluded by a constraint that is inactive at convergence. Yet the point found at convergence remains a stationary point whether or not the inactive constraints are included. Because an inactive $h^{(i)}$ has negative value, then the solution to $\min_{\mathbf{x}} \max_{\boldsymbol{\lambda}} \max_{\boldsymbol{\alpha}, \boldsymbol{\alpha} \geq 0} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\alpha})$ will have $\alpha_i = 0$. We can thus observe that at the solution, $\boldsymbol{\alpha} \odot \mathbf{h}(\mathbf{x}) = \mathbf{0}$. In other words, for all i , we know that at least one of the constraints $\alpha_i \geq 0$ or $h^{(i)}(\mathbf{x}) \leq 0$ must be active at the solution. To gain some intuition for this idea, we can say that either the solution is on the boundary imposed by the inequality and we must use its KKT multiplier to influence the solution to $\mathbf{x}$, or the inequality has no influence on the solution and we represent this by zeroing out its KKT multiplier.

A simple set of properties describe the optimal points of constrained optimization problems. These properties are called the Karush-Kuhn-Tucker (KKT) conditions (Karush, 1939; Kuhn and Tucker, 1951). They are necessary conditions, but not always sufficient conditions, for a point to be optimal. The conditions are:

- The gradient of the generalized Lagrangian is zero.
- All constraints on both $\mathbf{x}$ and the KKT multipliers are satisfied.
- The inequality constraints exhibit “complementary slackness”: $\boldsymbol{\alpha} \odot \mathbf{h}(\mathbf{x}) = \mathbf{0}$.

For more information about the KKT approach, see Nocedal and Wright (2006).